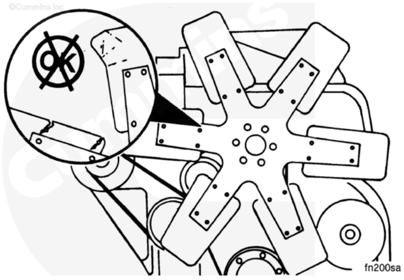
Trainings

Initial Check

⚠ WARNING ⚠

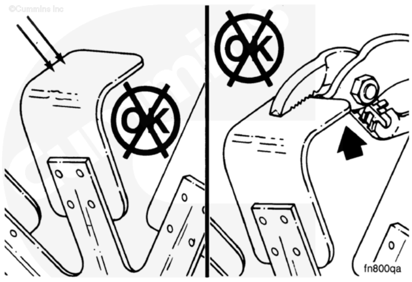
The cooling fan will engage when the engine is started. To reduce the possibility of personal injury, do not put your hands in the path of the rotating fan.



Check the fan for cracked, bent, or broken blades, and loose or worn rivets.

⚠ WARNING ⚠

Do not straighten a bent fan blade or continue to use a damage fan. A bent or damaged fan blade can fail during operation and cause personal injury or property damage.

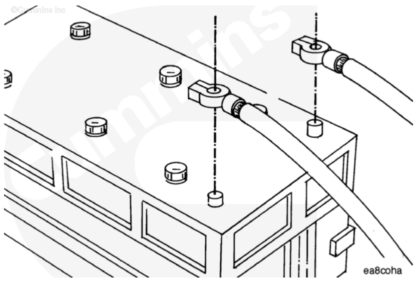


Do **not** straighten bent fan blades. Replace if necessary.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



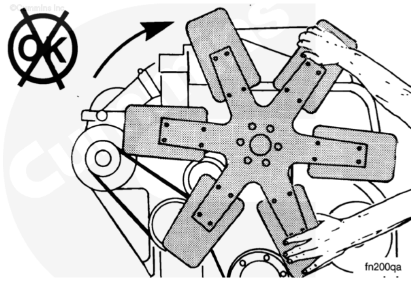
Disconnect the batteries or air supply line to the air starter (if equipped) to prevent accidental engine starting.

Remove

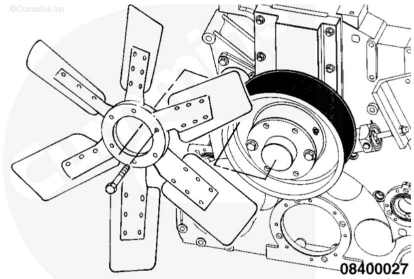
⚠ CAUTION ⚠

Do not pry or pull on the fan blades. The fan blades can be damaged.

Do **not** remove cooling fan by prying or pulling on fan blades.

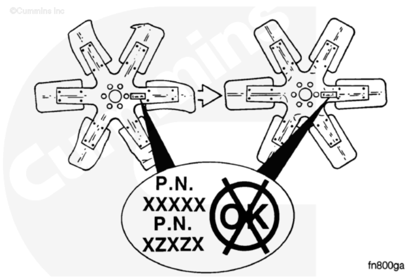


Remove the fan and spacer. Some engine applications will **not** use a fan spacer.



Install

Replace the original equipment fan with a fan of identical part number. Cummins Inc. **must** approve any fan changes.

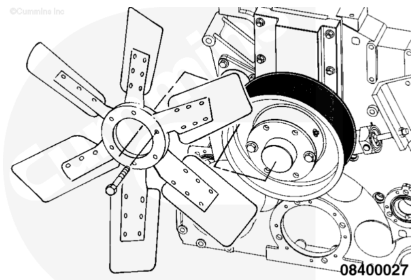


Install the fan and applicable spacer.

Tighten the mounting capscrews.

Torque Value: 135 n•m [100 ft-lb]

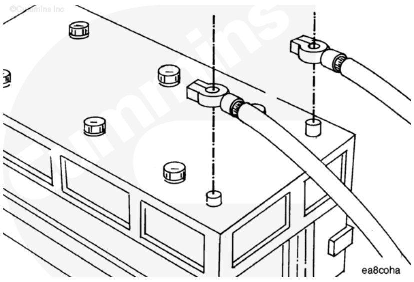
A minimum of 19.05 mm [3/4 in] of threads **must** be engaged in the fan hub.



Finishing Steps

⚠ WARNING ⚠

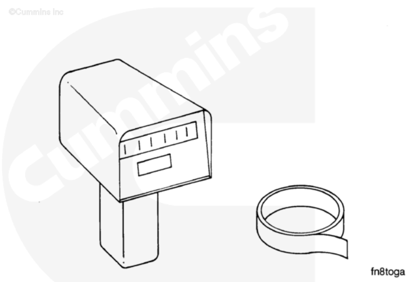
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Connect the batteries or air starter (if equipped) supply line.

Rotation Check

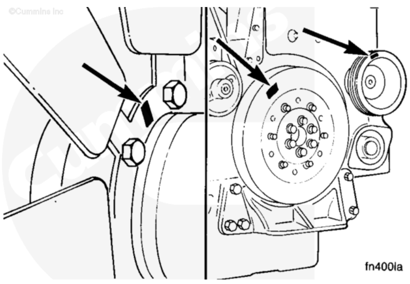
Use an optical tachometer, Part Number 3377462, or a strobe light to measure the fan rpm.



Mark a spot on the fan so the fan rpm can be measured.

Mark another spot on the vibration damper or accessory drive pulley to measure engine rpm.

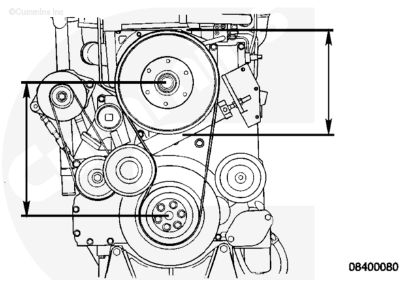
When using the optical tachometer, Part Number 3377462, use a piece of reflective tape, Part Number 3377464, to mark the spot.



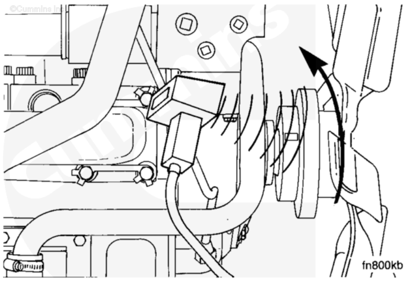
Measure the diameter of the fan belt pulley to determine the drive ratio.

Belt Driven Fan	
Fan Pulley Diameter	**Fan (rpm)**
311 mm [12.2 in]	0.70 x engine rpm

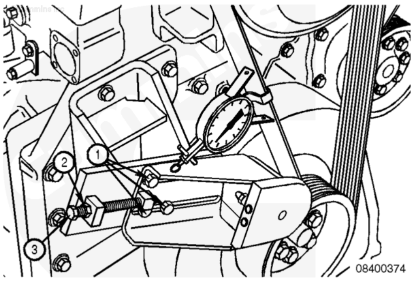
The fan center distance is the distance between the center of the fan hub and the center of the crankshaft.



Operate the engine. Read and compare the fan rpm to the engine rpm. Compare the fan rpm to the engine specification.



If the fan rpm is **not** correct, check the belt tension. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/89/89-008-002.html).



Last Modified: 15-May-2003